

JEE (MAIN+ADVANCED)

Mathematics

ENTHUSIAST COURSE



STUDY MATERIAL

Application of Derivative

English Medium

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RATE MEASURE, TANGENT & NORMAL

1. RATE MEASUREMENT :

Whenever one quantity y varies with another quantity x , satisfying some rule $y = f(x)$, then

$\frac{dy}{dx}$ (or $f'(x)$) represents the rate of change of y with respect to x and $\left. \frac{dy}{dx} \right|_{x=a}$ (or $f'(a)$) represents the rate

of change of y with respect to x at $x = a$.

Illustration 1 : The volume of a cube is increasing at a rate of $9\text{cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 10cm ?

Solution : Let x be the length of side, V be the volume and S be the surface area of the cube. Then $V = x^3$ and $S = 6x^2$, where x is a function of time t .

$$\frac{dV}{dt} = 9\text{cm}^3/\text{s} = \frac{d}{dt}(x^3) = 3x^2 \frac{dx}{dt}$$

$$\Rightarrow \frac{dx}{dt} = \frac{3}{x^2}$$

$$\frac{dS}{dt} = \frac{d}{dt}(6x^2) = 12x \left(\frac{3}{x^2} \right) = \frac{36}{x}$$

$$\left. \frac{dS}{dt} \right|_{x=10\text{cm}} = 3.6 \text{ cm}^2/\text{s}.$$

Illustration 2 : x and y are the sides of two squares such that $y = x - x^2$. Find the rate of change of the area of the second square with respect to the first square.

Solution : Given x and y are sides of two squares. Thus the area of two squares are x^2 and y^2

$$\text{We have to obtain } \frac{d(y^2)}{d(x^2)} = \frac{2y \frac{dy}{dx}}{2x} = \frac{y}{x} \cdot \frac{dy}{dx} \quad \dots\dots\dots (i)$$

$$\text{where the given curve is, } y = x - x^2 \Rightarrow \frac{dy}{dx} = 1 - 2x \quad \dots\dots\dots (ii)$$

$$\text{Thus, } \frac{d(y^2)}{d(x^2)} = \frac{y}{x} (1 - 2x) \quad [\text{from (i) and (ii)}]$$

$$\text{or } \frac{d(y^2)}{d(x^2)} = \frac{(x - x^2)(1 - 2x)}{x} \Rightarrow \frac{d(y^2)}{d(x^2)} = (2x^2 - 3x + 1)$$

The rate of change of the area of second square with respect to first square is $(2x^2 - 3x + 1)$

Do yourself - 1 :

- What is the rate of change of the area of a circle with respect to its radius r at $r = 6$ cm.
- A stone is dropped into a quiet lake and waves move in circles at the speed of 5 cm/s. At the instant when the radius of the circular wave is 8 cm, how fast is the enclosed area increasing ?
- How fast the area of a circle increases when its radius is 5 cm;
 - with respect to radius
 - with respect to diameter
- If area of circle increases at a rate of $2\text{ cm}^2/\text{sec}$, then find the rate at which area of the inscribed square increases.
- The volume of a cube is increasing at a rate of $7\text{ cm}^3/\text{sec}$. How fast is the surface area increasing when the length of an edge is 4 cm ?
- Sand is pouring from pipe at the rate of $12\text{ cm}^3/\text{s}$. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-sixth of radius of base. How fast is the height of the sand cone increasing when height is 4 cm ?
- The side of a square sheet is increasing at the rate of 4 cm per minute. The rate by which the area increasing when the side is 8 cm long is -
 (A) $16\text{ cm}^2/\text{min}$. (B) $32\text{ cm}^2/\text{min}$. (C) $64\text{ cm}^2/\text{min}$. (D) $80\text{ cm}^2/\text{min}$.
- The radius of an air bubble is increasing at the rate of 0.5 cm/sec. The rate by which the volume of the bubble is increasing when the radius is 1 cm, is -
 (A) $2\pi\text{ cm}^3/\text{s}$ (B) $\pi\text{ cm}^3/\text{s}$ (C) $4\pi\text{ cm}^3/\text{s}$ (D) $0.5\pi\text{ cm}^3/\text{s}$
- The point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the same rate is -
 (A) (4, 2) (B) (-4, -2) (C) (2, 4) (D) (-2, -4)
- A particle moves along the curve $y = x^2 + 2x$. Then the points on the curve where the x and y coordinates of the particle changing at the same rate, are
 (A) $\left(\frac{-3}{4}, \frac{-1}{2}\right)$ (B) $\left(\frac{-1}{2}, \frac{-3}{4}\right)$ (C) $\left(\frac{3}{4}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \frac{3}{4}\right)$

2. APPROXIMATION USING DIFFERENTIALS :

In order to calculate the approximate value of a function, differentials may be used where the differential of a function is equal to its derivative multiplied by the differential of the independent variable.

In general $dy = f'(x)dx$ or $df(x) = f'(x)dx$

Note :

- (i) For the independent variable 'x', increment Δx and differential dx are equal but this is not the case with the dependent variable 'y' i.e. $\Delta y \neq dy$.

$\therefore$ Approximate value of y when increment Δx is given to independent variable x in $y = f(x)$ is

$$y + \Delta y = f(x + \Delta x) = f(x) + \frac{dy}{dx} \cdot \Delta x$$

- (ii) The relation $dy = f'(x) dx$ can be written as $\frac{dy}{dx} = f'(x)$; thus the quotient of the differentials of 'y' and 'x' is equal to the derivative of 'y' w.r.t. 'x'.

Illustration 3 : Find the approximate value of square root of 25.2.

Solution : Let $f(x) = \sqrt{x}$

$$\text{Now, } f(x + \Delta x) - f(x) = f'(x) \cdot \Delta x = \frac{\Delta x}{2\sqrt{x}}$$

we may write, $25.2 = 25 + 0.2$

Taking $x = 25$ and $\Delta x = 0.2$, we have

$$f(25.2) - f(25) = \frac{0.2}{2\sqrt{25}}$$

$$\text{or } f(25.2) - \sqrt{25} = \frac{0.2}{10} = 0.02 \Rightarrow f(25.2) = 5.02$$

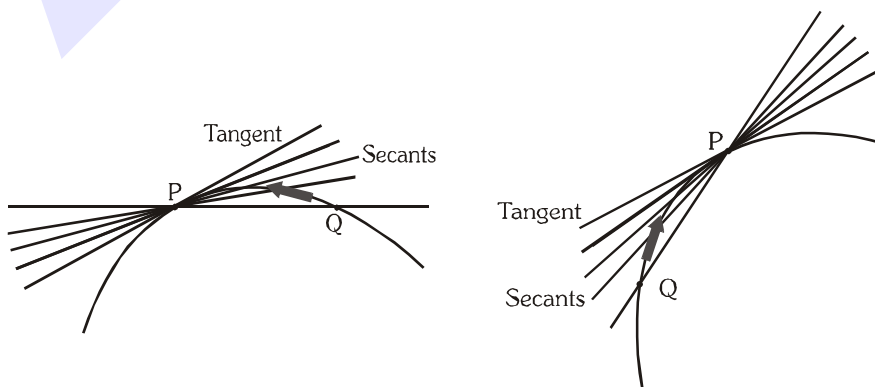
$$\text{or } \sqrt{(25.2)} = 5.02$$

Do yourself - 2 :

- Find the approximate value of $(0.009)^{1/3}$.
- Find the approximate value of $\sqrt{49.5}$.
- Find the approximate value of $(32.15)^{1/5}$.
- Find the approximate value of $f(5.001)$, where $f(x) = x^3 - 7x^2 + 15$.
(A) -34.995 (B) -33.995 (C) 33.995 (D) 34.995
- Use differentials to approximate the values of ; (a) $\sqrt{36.6}$ and (b) $\sqrt[3]{26}$.
- If the radius of a sphere is measured as 9 cm with an error of 0.03 cm, then find the approximate error in calculating its volume.

3. TANGENT TO THE CURVE AT A POINT :

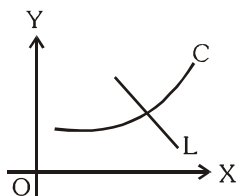
The tangent to the curve at 'P' is the line through P whose slope is limit of the secant slopes as $Q \rightarrow P$ from either side.



4. MYTHS ABOUT TANGENT :

- (a) **Myth :** A line meeting the curve only at one point is a tangent to the curve.

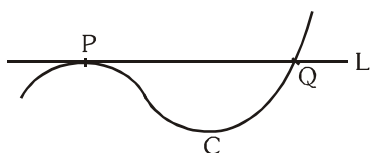
Explanation : A line meeting the curve in one point is not necessarily tangent to it.



Here L is not tangent to C

- (b) **Myth :** A line meeting the curve at more than one point is not a tangent to the curve.

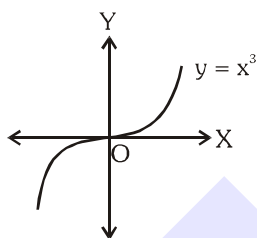
Explanation : A line may meet the curve at several points and may still be tangent to it at some point



Here L is tangent to C at P, and cutting it again at Q.

- (c) **Myth :** Tangent at a point to the curve can not cross it at the same point.

Explanation : A line may be tangent to the curve and also cross it.



Here X-axis is tangent to $y = x^3$ at origin.

5. NORMAL TO THE CURVE AT A POINT :

A line which is perpendicular to the tangent at the point of contact is called normal to the curve at that point.

6. POINTS TO REMEMBER :

- (a) The value of the derivative at $P(x_1, y_1)$ gives the slope of the tangent to the curve at P. Symbolically

$$f'(x_1) = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} = \text{Slope of tangent at } P(x_1, y_1) = m(\text{say}).$$

- (b) Equation of tangent at (x_1, y_1) is ; $y - y_1 = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} (x - x_1)$

- (c) Equation of normal at (x_1, y_1) is ; $y - y_1 = -\left. \frac{1}{\frac{dy}{dx}} \right|_{(x_1, y_1)} (x - x_1).$

Note :

- (i) The point $P(x_1, y_1)$ will satisfy the equation of the curve & the equation of tangent & normal line.
- (ii) If the tangent at any point P on the curve is parallel to the axis of x then $dy/dx = 0$ at the point P .
- (iii) If the tangent at any point on the curve is parallel to the axis of y , then dy/dx not defined or $dx/dy = 0$.
- (iv) If the tangent at any point on the curve is equally inclined to both the axes then $dy/dx = \pm 1$.
- (v) If a curve passing through the origin be given by a rational integral algebraic equation, then the equation of the tangent (or tangents) at the origin is obtained by equating to zero the terms of the lowest degree in the equation. e.g. If the equation of a curve be $x^2 - y^2 + x^3 + 3x^2y - y^3 = 0$, the tangents at the origin are given by $x^2 - y^2 = 0$ i.e. $x + y = 0$ and $x - y = 0$

Illustration 4: Find the equation of the tangent to the curve $y = (x^3 - 1)(x - 2)$ at the points where the curve cuts the x-axis.

Solution : The equation of the curve is $y = (x^3 - 1)(x - 2)$ (i)

It cuts x-axis at $y = 0$. So, putting $y = 0$ in (i), we get $(x^3 - 1)(x - 2) = 0$

$$\Rightarrow (x-1)(x-2)(x^2+x+1)=0 \Rightarrow x-1=0, x-2=0 \quad \left[\because x^2+x+1 \neq 0 \right]$$

$$\Rightarrow x = 1, 2.$$

Thus, the points of intersection of curve (i) with x-axis are (1, 0) and (2, 0). Now,

$$y = (x^3 - 1)(x - 2) \Rightarrow \frac{dy}{dx} = 3x^2(x - 2) + (x^3 - 1) \Rightarrow \left(\frac{dy}{dx}\right)_{(1,0)} = -3 \text{ and } \left(\frac{dy}{dx}\right)_{(2,0)} = 7$$

The equations of the tangents at $(1, 0)$ and $(2, 0)$ are respectively

$$y-0=-3(x-1) \text{ and } y-0=7(x-2) \Rightarrow y+3x-3=0 \text{ and } 7x-y-14=0 \quad \text{Ans.}$$

Illustration 5 : The equation of the normal to the curve $y = x + \sin x \cos x$ at $x = \frac{\pi}{2}$ is -

- (A) $x = 2$ (B) $x = \pi$ (C) $x + \pi = 0$ (D) $2x = \pi$

Solution : $\because x = \frac{\pi}{2} \Rightarrow y = \frac{\pi}{2} + 0 = \frac{\pi}{2}$, so the given point = $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

Now from the given equation $\frac{dy}{dx} = 1 + \cos^2 x - \sin^2 x \Rightarrow \left(\frac{dy}{dx}\right)_{\left(\frac{\pi}{2}, \frac{\pi}{2}\right)} = 1 + 0 - 1 = 0$

$\Rightarrow$ The curve has vertical normal at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$.

The equation to this normal is $x = \frac{\pi}{2}$

$$\Rightarrow x - \frac{\pi}{2} = 0 \Rightarrow 2x = \pi$$

Ans. (D)

Illustration 6 : The equation of normal to the curve $x + y = x^y$, where it cuts x-axis is -

- (A) $y = x + 1$ (B) $y = -x + 1$ (C) $y = x - 1$ (D) $y = -x - 1$

Solution : Given curve is $x + y = x^y$ (i)

at x-axis $y = 0$,

$$\therefore x + 0 = x^0 \Rightarrow x = 1$$

$\therefore$ Point is A(1, 0)

Now to differentiate $x + y = x^y$ take log on both sides

$$\Rightarrow \log(x + y) = y \log x \quad \therefore \frac{1}{x + y} \left\{ 1 + \frac{dy}{dx} \right\} = y \cdot \frac{1}{x} + (\log x) \frac{dy}{dx}$$

$$\text{Putting } x = 1, y = 0 \quad \left\{ 1 + \frac{dy}{dx} \right\} = 0 \Rightarrow \left(\frac{dy}{dx} \right)_{(1,0)} = -1$$

$\therefore$ slope of normal = 1

$$\text{Equation of normal is, } \frac{y - 0}{x - 1} = 1 \Rightarrow y = x - 1$$

Ans. (C)

Illustration 7 : Find the slope of normal at the point with abscissa $x = -2$ of the graph of the function $f(x) = |x^2 - |x||$

Solution : At $x = -2$, $f(x)$ becomes

$$f(x) = x^2 + x$$

$$\frac{dy}{dx} = 2x + 1 = -3$$

$$\text{Slope of normal} = \frac{1}{3}$$

Illustration 8 : If $y = 4x - 5$ is a tangent to the curve $y^2 = px^3 + q$ at (2, 3), then

- (A) $p = 2, q = -7$ (B) $p = -2, q = 7$
(C) $p = -2, q = -7$ (D) $p = 2, q = 7$

Solution : $\frac{dy}{dx} = 4$ & $9 = 8p + q$

$$2y \frac{dy}{dx} = 3px^2$$

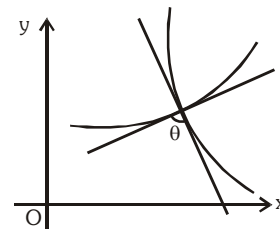
$$6 \frac{dy}{dx} = 3p(4) \Rightarrow \frac{dy}{dx} = 2p = 4 \Rightarrow p = 2 \text{ \& } q = -7$$

Do yourself - 3 :

- Find the distance between the point (1,1) and the tangent to the curve $y = e^{2x} + x^2$ drawn at the point where the curve cuts y-axis.
- Find the equation of a line passing through (-2,3) and parallel to tangent at origin for the circle $x^2 + y^2 + x - y = 0$.
- If the tangent at a point P, with parameter t, on the curve $x = 4t^2 + 3, y = 8t^3 - 1, t \in \mathbb{R}$, meets the curve again at a point Q, then the coordinates of Q are :
 (A) $(t^2 + 3, t^3 - 1)$ (B) $(t^2 + 3, -t^3 - 1)$
 (C) $(16t^2 + 3, -64t^3 - 1)$ (D) $(4t^2 + 3, -8t^3 - 1)$
- Find the equation of tangent and normal to the curve $x = \frac{2at^2}{1+t^2}, y = \frac{2at^3}{1+t^2}$ at the point $t = \frac{1}{2}$.
- The points on the curve $y = 12x - x^3$ at which the gradient is zero are
 (A) (0, 2), (2, 16) (B) (0, -2), (2, -16)
 (C) (2, -16), (-2, 16) (D) (2, 16), (-2, -16)
- The area of the triangle formed by the coordinate axes and a tangent to the curve $xy = a^2$ at the point (x_1, y_1) on it is
 (A) $\frac{a^2 x_1}{y_1}$ (B) $\frac{a^2 y_1}{x_1}$ (C) $2a^2$ (D) $4a^2$
- The point of the curve $y^2 = 2(x - 3)$ at which the normal is parallel to the line $y - 2x + 1 = 0$ is
 (A) (5, 2) (B) $\left(-\frac{1}{2}, -2\right)$ (C) (5, -2) (D) $\left(\frac{3}{2}, 2\right)$
- The line $x + y = 2$ is tangent to the curve $x^2 = 3 - 2y$ at its point
 (A) (1, 1) (B) (-1, 1) (C) $(\sqrt{3}, 0)$ (D) (3, -3)
- At what point on the curve $x^3 - 8a^2 y = 0$, the slope of the normal is $-\frac{2}{3}$
 (A) (a, a) (B) (2a, -a) (C) (2a, a) (D) None of these
- At which point the line $\frac{x}{a} + \frac{y}{b} = 1$, touches the curve $y = be^{-x/a}$
 (A) (0, 0) (B) (0, a) (C) (0, b) (D) (b, 0)
- If normal to the curve $y = f(x)$ is parallel to x-axis, then correct statement is
 (A) $\frac{dy}{dx} = 0$ (B) $\frac{dy}{dx} = 1$ (C) $\frac{dx}{dy} = 0$ (D) None of these
- The abscissae of the points, where the tangent to curve $y = x^3 - 3x^2 - 9x + 5$ is parallel to x-axis, are
 (A) 0 and 0 (B) $x = 1$ and -1 (C) $x = 1$ and -3 (D) $x = -1$ and 3

7. ANGLE OF INTERSECTION BETWEEN TWO CURVES :

Angle of intersection between two curves is defined as the angle between the two tangents drawn to the two curves at their point of intersection.

**Orthogonal curves :**

If the angle between two curves at each point of intersection is 90° then they are called **orthogonal curves**.

For example, the curves $x^2 + y^2 = r^2$ & $y = mx$ are orthogonal curves.

Illustration 9 : The angle of intersection between the curve $x^2 = 32y$ and $y^2 = 4x$ at point $(16, 8)$ is-

- (A) 60° (B) 90° (C) $\tan^{-1}\left(\frac{3}{5}\right)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$

Solution :

$$x^2 = 32y \Rightarrow \frac{dy}{dx} = \frac{x}{16} \Rightarrow y^2 = 4x \Rightarrow \frac{dy}{dx} = \frac{2}{y}$$

$$\therefore \text{ at } (16, 8), \left(\frac{dy}{dx}\right)_1 = 1, \left(\frac{dy}{dx}\right)_2 = \frac{1}{4}$$

$$\text{So required angle} = \tan^{-1} \left(\frac{1 - \frac{1}{4}}{1 + 1 \left(\frac{1}{4}\right)} \right) = \tan^{-1} \left(\frac{3}{5} \right)$$

Ans. (C)

Illustration 10 : Check the orthogonality of the curves $y^2 = x$ & $x^2 = y$.

Solution : Solving the curves simultaneously we get points of intersection as $(1, 1)$ and $(0, 0)$.

$$\text{At } (1,1) \text{ for first curve } 2y \left(\frac{dy}{dx}\right)_1 = 1 \Rightarrow m_1 = \frac{1}{2}$$

$$\& \text{ for second curve } 2x = \left(\frac{dy}{dx}\right)_2 \Rightarrow m_2 = 2$$

$$m_1 m_2 \neq -1 \text{ at } (1,1).$$

But at $(0,0)$ clearly x -axis & y -axis are their respective tangents hence they are orthogonal at $(0,0)$ but not at $(1,1)$. Hence these curves are not said to be orthogonal.

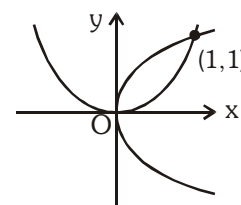


Illustration 11 : If curve $y = 1 - ax^2$ and $y = x^2$ intersect orthogonally then the value of a is -

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) 2 (D) 3

Solution : $y = 1 - ax^2 \Rightarrow \frac{dy}{dx} = -2ax$ $y = x^2 \Rightarrow \frac{dy}{dx} = 2x$

Two curves intersect orthogonally if $\left(\frac{dy}{dx}\right)_1 \left(\frac{dy}{dx}\right)_2 = -1$

$\Rightarrow (-2ax)(2x) = -1 \quad \Rightarrow 4ax^2 = 1 \quad \dots (i)$

Now eliminating y from the given equations we have $1 - ax^2 = x^2$

$\Rightarrow (1 + a)x^2 = 1 \quad \dots (ii)$

Eliminating x^2 from (i) and (ii) we get $\frac{4a}{1+a} = 1 \quad \Rightarrow a = \frac{1}{3}$

Ans. (B)

Do yourself -4 :

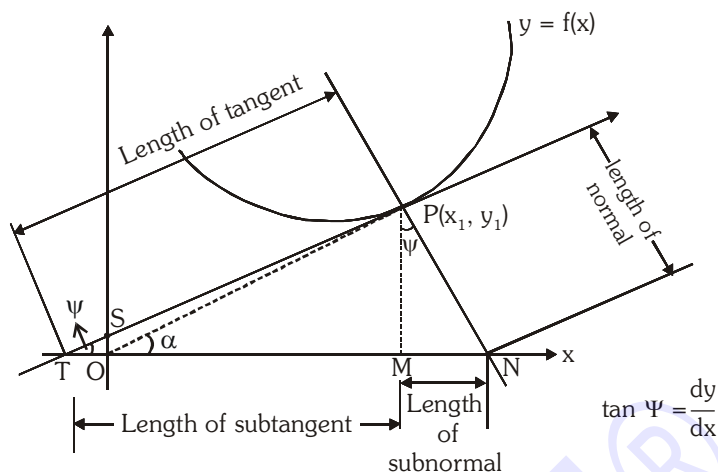
1. If two curves $y = a^x$ and $y = b^x$ intersect at an angle α , then find the value of $\tan \alpha$.
2. Find the angle of intersection of curves $y = 4 - x^2$ and $y = x^2$.
3. The angle of intersection between the curve $x^2 = 32y$ and $y^2 = 4x$ at point (16,8) is

- (A) 60° (B) 90° (C) $\tan^{-1}\left(\frac{3}{5}\right)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$

4. Check the orthogonality of the curves $y^2 = x$ & $x^2 = y$.
5. If curve $y = 1 - ax^2$ and $y = x^2$ intersect orthogonally then the value of a is

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) 2 (D) 3

8. LENGTH OF TANGENT, SUBTANGENT, NORMAL & SUBNORMAL :



(a) Length of the tangent (PT) = $\frac{y_1 \sqrt{1 + [f'(x_1)]^2}}{f'(x_1)}$

(b) Length of Subtangent (MT) = $\left| \frac{y_1}{f'(x_1)} \right|$

(c) Length of Normal (PN) = $\left| y_1 \sqrt{1 + \left[f'(x_1) \right]^2} \right|$

(d) Length of Subnormal (MN) = $|y_1 f'(x_1)|$

(e) **Initial ordinate** : Y intercept of tangent at point $P(x, y) = OS$

(f) **Radius vector (Polar radius) :** Line segment joining origin to point $P(x, y) = OP$

(g) **Vectorial angle :** Angle made by radius vector with positive direction of x-axis in anticlockwise direction is called vectorial angle. In given figure α is vectorial angle.

Illustration 12 : The length of the normal to the curve $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ at $\theta = \frac{\pi}{2}$ is -

(A) 2a

(B) $\frac{a}{2}$

(C) $\sqrt{2}a$

(D) $\frac{a}{\sqrt{2}}$

Solution :

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{a \sin \theta}{a(1 + \cos \theta)} = \tan \frac{\theta}{2} \Rightarrow \left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{2}} = \tan\left(\frac{\pi}{4}\right) = 1$$

Also at $\theta = \frac{\pi}{2}$, $y = a \left(1 - \cos \frac{\pi}{2} \right) = a$

$$\therefore \text{required length of normal} = y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = a \sqrt{1+1} = \sqrt{2}a$$

Ans. (C)

Illustration 13 : The length of the tangent to the curve $x = a \left(\cos t + \log \tan \frac{t}{2} \right)$, $y = a \sin t$ is

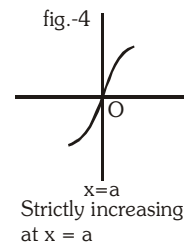
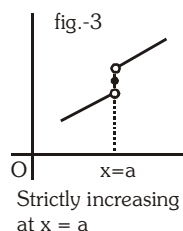
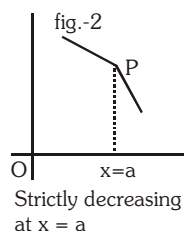
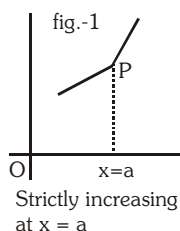
- (A) ax (B) ay (C) a (D) xy

Solution : $\frac{dy}{dx} = \left(\frac{dy}{dt} \right) / \left(\frac{dx}{dt} \right) = \frac{a \cos t}{a \left(-\sin t + \frac{1}{\sin t} \right)} = \tan t$

$$\therefore \text{length of the tangent} = y \frac{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{\left(\frac{dy}{dx}\right)} = a \sin t \frac{\sqrt{1 + \tan^2 t}}{\tan t} = a \sin t \left(\frac{\sec t}{\tan t}\right) = a \text{ Ans. (C)}$$

Do yourself - 5 :

1. Prove that at any point of a curve, the product of the length of sub tangent and the length of sub normal is equal to square of the ordinates of point of contact.
2. At any point of a curve $\sqrt{\frac{\text{Length of subnormal}}{\text{Length of subtangent}}}$ is equal to -
 - (A) the abscissa of that point
 - (B) the ordinate of that point
 - (C) slope of the tangent at that point
 - (D) slope of the normal at that point
3. The length of normal to the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ at the point $\theta = \pi/2$ is
 - (A) $2a$
 - (B) $a/2$
 - (C) $\sqrt{2}a$
 - (D) $\frac{a}{\sqrt{2}}$
4. The length of the normal at point 't' of the curve $x = a(t + \sin t)$, $y = a(1 - \cos t)$ is
 - (A) $a \sin t$
 - (B) $2a \sin^3(t/2) \sec(t/2)$
 - (C) $2a \sin(t/2) \tan(t/2)$
 - (D) $2a \sin(t/2)$
5. If ST and SN are the lengths of the subtangent and the subnormal at the point $\theta = \frac{\pi}{2}$ on the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, $a \neq 1$, then
 - (A) $ST = SN$
 - (B) $ST = 2SN$
 - (C) $ST^2 = aSN^3$
 - (D) $ST^3 = aSN$
6. For the curve $xy = c^2$ the subnormal at any point varies as
 - (A) x^2
 - (B) x^3
 - (C) y^2
 - (D) y^3



2. INCREASING & DECREASING NATURE OF A FUNCTION OVER AN INTERVAL :

Consider an interval $I \subseteq D_f$

I. Increasing Over an Interval I :

$\forall x_1, x_2 \in I, x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$, then f is increasing over interval I.

II. Decreasing Over an Interval I :

$\forall x_1, x_2 \in I, x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$, then f is decreasing over interval I.

III. Strictly increasing over an Interval I :

$\forall x_1, x_2 \in I, x_1 < x_2 \Leftrightarrow f(x_1) < f(x_2)$, then f is strictly increasing over interval I.

IV. Strictly decreasing over an Interval I :

$\forall x_1, x_2 \in I, x_1 < x_2 \Leftrightarrow f(x_1) > f(x_2)$, then f is strictly decreasing over interval I.

3. MONOTONIC FUNCTION :

If a function is either increasing or decreasing over an interval then it is said to be monotonic function over the interval.

If a function is either strictly increasing or strictly decreasing over an interval then it is said to be strictly monotonic function over the interval.

4. FOR DIFFERENTIABLE FUNCTIONS :

Consider an interval $I (\subseteq D_f)$ that can be $[a, b]$ or (a, b) or $[a, b)$ or $(a, b]$.

(1) $f'(x) > 0 \forall x \in I \Rightarrow f$ is strictly increasing function over the interval I.

(2) $f'(x) \geq 0 \forall x \in I \Rightarrow f$ is increasing function over the interval I.

(3) $f'(x) \geq 0 \forall x \in I$ and $f'(x) = 0$ do not form any interval (that means $f'(x) = 0$ at discrete points) $\Rightarrow f$ is strictly increasing function over the interval I.

(4) $f'(x) < 0 \forall x \in I \Rightarrow f$ is strictly decreasing function over the interval I.

(5) $f'(x) \leq 0 \forall x \in I \Rightarrow f$ is decreasing function over the interval I.

(6) $f'(x) \leq 0 \forall x \in I$ and $f'(x) = 0$ do not form any interval (that means $f'(x) = 0$ at discrete points) $\Rightarrow f$ is strictly decreasing function over the interval I.

Illustration 1 : Let $f(x) = x^3 - 3x + 2$. Examine the monotonicity of function at points $x = 0, 1$ & 2 .

Solution :

at $x = 0$,

$$f(0) = 2, f(0 + h) = h(h - 3) + 2 < 2$$

$$f(0 - h) = h(3 - h) + 2 > 2$$

$$\Rightarrow f(0 - h) > f(0) > f(0 + h)$$

f is decreasing or strictly decreasing at $x = 0$.

at $x = 1$, $f(1 - h) > f(1) < f(1 + h)$ neither increasing nor decreasing

Similarly at $x = 2$, $f(2 - h) < f(2) < f(2 + h)$ increasing at $x = 2$.

Illustration 2 : Prove that the function $f(x) = \log(x^3 + \sqrt{x^6 + 1})$ is strictly increasing.

Solution : Now, $f(x) = \log(x^3 + \sqrt{x^6 + 1})$

$$f'(x) = \frac{1}{x^3 + \sqrt{x^6 + 1}} \left(3x^2 + \frac{6x^5}{2\sqrt{x^6 + 1}} \right) = \frac{3x^2}{\sqrt{x^6 + 1}} \geq 0$$

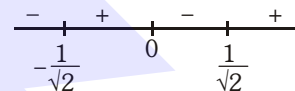
$\Rightarrow f(x)$ is strictly increasing.

Illustration 3 : Find the intervals of monotonicity of the function $y = x^2 - \log_e |x|$, ($x \neq 0$).

Solution : Let $y = f(x) = x^2 - \log_e |x|$

$$f'(x) = 2x - \frac{1}{x} ; \text{ for all } x (x \neq 0)$$

$$f'(x) = \frac{2x^2 - 1}{x} \Rightarrow f'(x) = \frac{(\sqrt{2}x - 1)(\sqrt{2}x + 1)}{x}$$



So $f'(x) \geq 0$ when $x \in \left[-\frac{1}{\sqrt{2}}, 0\right) \cup \left[\frac{1}{\sqrt{2}}, \infty\right)$ and $f'(x) \leq 0$

when $x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right] \cup \left(0, \frac{1}{\sqrt{2}}\right]$

$\therefore f(x)$ is strictly increasing when $x \in \left[-\frac{1}{\sqrt{2}}, 0\right); \left[\frac{1}{\sqrt{2}}, \infty\right)$

and strictly decreasing when $x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right]; \left(0, \frac{1}{\sqrt{2}}\right]$

Illustration 4 : If $g(x) = f(x) + f(1 - x)$ and $f''(x) < 0$; $0 \leq x \leq 1$, show that $g(x)$ strictly increasing in $x \in (0, 1/2)$ and strictly decreasing in $x \in (1/2, 1)$

Solution : $\therefore f''(x) < 0 \Rightarrow f'(x)$ is strictly decreasing function.

$$\text{Now, } g(x) = f(x) + f(1 - x) \quad \therefore g'(x) = f'(x) - f'(1 - x) \quad \dots\dots\dots (i)$$

Case I : If $x > (1 - x) \Rightarrow x > 1/2$

$$\therefore f'(x) < f'(1 - x) \Rightarrow f'(x) - f'(1 - x) < 0$$

$$\Rightarrow g'(x) < 0$$

$\therefore g(x)$ strictly decreases in $x \in \left(\frac{1}{2}, 1\right)$

Case II : If $x < (1 - x) \Rightarrow x < 1/2$

$$\therefore f'(x) > f'(1 - x) \Rightarrow f'(x) - f'(1 - x) > 0$$

$$\Rightarrow g'(x) > 0$$

$\therefore g(x)$ strictly increases in $x \in (0, 1/2)$

Illustration 5 : Which of the following functions are strictly decreasing on $\left(0, \frac{\pi}{2}\right)$

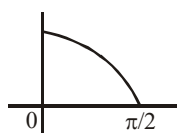
(A) $\cos x$

(B) $\cos 2x$

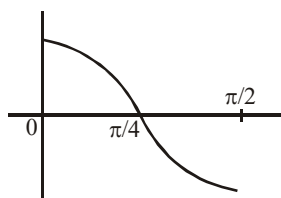
(C) $\cos 3x$

(D) $\tan x$

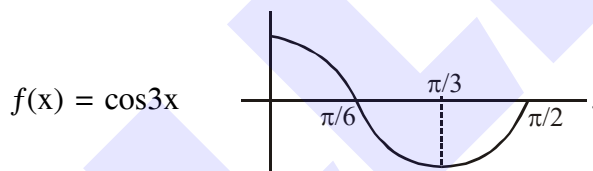
Solution :



$$f(x) = \cos x$$



$$f(x) = \cos 2x$$



$$f(x) = \cos 3x$$

Non monotonic

$f(x) = \tan x$ is strictly increasing in $\left(0, \frac{\pi}{2}\right)$

Option A & B are correct.

Illustration 6 : Prove that the equation $e^{(x-1)} + x = 2$ has one solution.

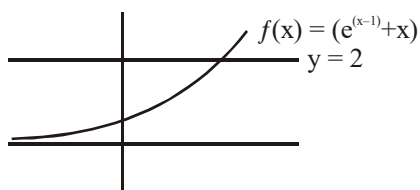
Solution :

$$\text{Let } f(x) = e^{(x-1)} + x$$

$$f'(x) = e^{(x-1)} + 1$$

$f(x)$ is strictly increasing function

$$\lim_{x \rightarrow \infty} f(x) = \infty \quad \& \quad \lim_{x \rightarrow -\infty} f(x) = 0$$

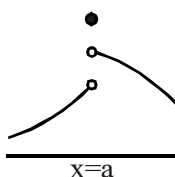


$f(x) = 2$ has exactly one solution.

Do yourself - 1 :

- If function $f(x) = x^3 + \lambda x^2 - \lambda x + 1$ is increasing at $x = 0$ & decreasing at $x = 1$, then find the greatest integral value of λ .
- If $f(x) = \sin x + \ln |\sec x + \tan x| - 2x$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then check the monotonicity of $f(x)$
- Prove that $y = e^x + \sin x$ is strictly increasing in $x \in \mathbb{R}^+$
- Let $f(x) = x^3 - 3x + 2$. Examine the monotonicity of function at points $x = 0, 1$ & 2 .
- Prove that the function $f(x) = \log(x^3 + \sqrt{x^6 + 1})$ is strictly increasing.
- The function f , defined by $f(x) = (x+2)e^{-x}$ is -
 (A) decreasing for all x
 (B) decreasing in $(-\infty, -1)$ and increasing in $(-1, \infty)$
 (C) increasing for all x
 (D) decreasing in $(-1, \infty)$ and increasing in $(-\infty, -1)$
- Function $f(x) = x^3 + 6x^2 + (9+2k)x + 1$ is increasing function if
 (A) $k \geq \frac{3}{2}$ (B) $k > \frac{3}{2}$ (C) $k < \frac{3}{2}$ (D) $k \leq \frac{3}{2}$
- The complete set of values of 'a' for which the function $f(x) = \sin x - \cos x - ax + b$ decreases for all real values of x , is
 (A) $a \geq -\sqrt{2}$ (B) $a \leq -\sqrt{2}$ (C) $a \leq \sqrt{2}$ (D) $a \geq \sqrt{2}$
- $f(x) = |x-1| + 2|x-3| - |x+2|$ is monotonically increasing at $x = ?$
 (A) -2 (B) 0 (C) 3 (D) 5
- Let $f(x) = \begin{cases} x & x \in [1, 2] \\ 5-x & x \in (2, 4) \\ 2 & x = 4 \\ 7-x & x \in (4, 6] \end{cases}$ then which of the following statement is/are correct about $f(x)$?
 (A) Function is strictly increasing at point $x = 2$
 (B) Function is strictly increasing at point $x = 4$
 (C) Function is not increasing at point $x = 2$ and $x = 4$
 (D) Function is not increasing at point $x = 1$

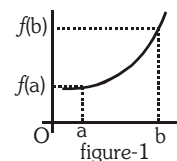
11. In the following graph comment on monotonicity of $f(x)$ at $x = a$.



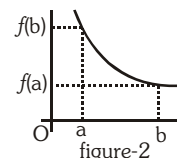
- (A) Monotonically increasing at $x = a$
 (B) Monotonically decreasing at $x = a$
 (C) neither monotonically increasing nor decreasing at $x = a$
 (D) Can't say at $x = a$
12. Function $f(x) = \ln \sin x$ is monotonically increasing when -
 (A) $x \in (\pi/2, \pi)$ (B) $x \in (-\pi/2, 0)$ (C) $x \in (0, \pi)$ (D) $x \in (0, \pi/2)$
13. Which of the following statements is/are correct ?
 (A) $x + \sin x$ is increasing function
 (B) $\sec x$ is neither increasing nor decreasing function
 (C) $x + \sin x$ is decreasing function
 (D) $\sec x$ is an increasing function
14. If $f(x) = 2x + \cot^{-1} x + \ln(\sqrt{1+x^2} - x)$, then $f(x)$:
 (A) increases in $[0, \infty)$
 (B) decreases in $[0, \infty)$
 (C) neither increases nor decreases in $[0, \infty)$
 (D) increases in $(-\infty, \infty)$
15. Let $\phi(x) = (f(x))^3 - 3(f(x))^2 + 4f(x) + 5x + 3 \sin x + 4 \cos x \quad \forall x \in \mathbb{R}$, where $f(x)$ is a differentiable function $\forall x \in \mathbb{R}$, then
 (A) ϕ is increasing whenever f is increasing
 (B) ϕ is increasing whenever f is decreasing
 (C) ϕ is decreasing whenever f is decreasing
 (D) ϕ is decreasing if $f'(x) = -11$
16. Find the intervals of monotonicity of the functions in $[0, 2\pi]$
 (a) $f(x) = \sin x - \cos x$ in $x \in [0, 2\pi]$
 (b) $g(x) = 2 \sin x + \cos 2x$ in $(0 \leq x \leq 2\pi)$
 (c) $f(x) = \frac{4 \sin x - 2x - x \cos x}{2 + \cos x}$
17. Let $f(x) = 1 - x - x^3$. Find all real values of x satisfying the inequality, $1 - f(x) - f^3(x) > f(1 - 5x)$

5. GREATEST AND LEAST VALUE OF A FUNCTION :

- (a) If a continuous function $y = f(x)$ is increasing in the closed interval $[a, b]$, then $f(a)$ is the least value and $f(b)$ is the greatest value of $f(x)$ in $[a, b]$ (figure-1)



- (b) If a continuous function $y = f(x)$ is decreasing in $[a, b]$, then $f(b)$ is the least and $f(a)$ is the greatest value of $f(x)$ in $[a, b]$. (figure-2)



- (c) If a continuous function $y = f(x)$ is increasing/decreasing in the (a, b) , then no greatest and least value exist.

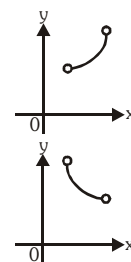


Illustration 7 : Show that $f(x) = \sin^{-1} \frac{x}{\sqrt{1+x^2}} - \ln x$ is strictly decreasing in $x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3} \right]$.

Also find its range.

Solution :
$$f(x) = \sin^{-1} \frac{x}{\sqrt{1+x^2}} - \ln x = \tan^{-1} x - \ln x \Rightarrow f'(x) = \frac{1}{1+x^2} - \frac{1}{x} = \frac{-(1+x^2-x)}{x(1+x^2)}$$

$$\therefore f'(x) \leq 0 \quad \forall x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3} \right]$$

$\Rightarrow f(x)$ is strictly decreasing.

$$f(x)|_{\max} = f\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} + \frac{1}{2} \ln 3 \quad \& \quad f(x)|_{\min} = f(\sqrt{3}) = \frac{\pi}{3} - \frac{1}{2} \ln 3$$

$$\therefore \text{Range of } f(x) = \left[\frac{\pi}{3} - \frac{1}{2} \ln 3, \frac{\pi}{6} + \frac{1}{2} \ln 3 \right]$$

Ans.

Illustration 8 : Find the greatest and least value of $f(x) = x^3 + 5x + e^x$ in $[1, 3]$

Solution : $f'(x) = 3x^2 + 5 + e^x \Rightarrow f(x)$ is strictly increasing.

$$\text{Least value} = f(1) = 6 + e$$

$$\text{greatest value} = f(3) = (42 + e^3)$$

6. PROVING INEQUALITIES USING MONOTONICITY :

Comparison of two functions $f(x)$ and $g(x)$ can be done by analysing their monotonic behaviour.

Illustration 9 : Let $f(x)$ and $g(x)$ are two function which are defined and differentiable for all $x \geq x_0$.
If $f(x_0) = g(x_0)$ and $f'(x) > g'(x)$ for all $x > x_0$ then

- (A) $f(x) < g(x)$ for some $x > x_0$ (B) $f(x) = g(x)$ for some $x > x_0$
(C) $f(x) > g(x)$ only for some $x > x_0$ (D) $f(x) > g(x)$ for all $x > x_0$

Solution : Let $h(x) = f(x) - g(x)$
 $h'(x) = f'(x) - g'(x) > 0 \quad \forall x > x_0$
 $h(x_0) = 0$
 $\Rightarrow h(x) > 0 \quad \forall x > x_0$
Ans. (D)

Do yourself - 2 :

- Let $f(x) = \frac{x^3}{3} - \frac{x^2}{2} + 2$ in $[-2, 2]$. Find the greatest and least value of $f(x)$ in $[-2, 2]$
- Prove that $x > \sin x > x - \frac{x^3}{6}$ for all $x > 0$.
- Show that $f(x) = \sin^{-1} \frac{x}{\sqrt{1+x^2}} - \ln x$ is strictly decreasing in $x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3} \right]$. Also find range of $f(x)$ for given x .
- For $x \in (0, 1)$ prove that $x - \frac{x^3}{3} < \tan^{-1} x < x - \frac{x^3}{6}$.
- The greatest, the least values of the function, $f(x) = 2 - \sqrt{1 + 2x + x^2}$, $x \in [-2, 1]$ are respectively
(A) 2, 1 (B) 2, -1 (C) 2, 0 (D) -2, 3
- Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let $m(b)$ be the minimum value of $f(x)$. As b varies, the range of $m(b)$ is
(A) $[0, 1]$ (B) $\left(0, \frac{1}{2}\right]$ (C) $\left[\frac{1}{2}, 1\right]$ (D) $(0, 1]$
- Which of the following is true
(A) $\sin x - \sin 2x \leq 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$ (B) $\sin x - \sin 2x \geq 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$
(C) $\sin x - \sin 2x \leq 2x$ for all $x \in \left[-\frac{\pi}{3}, 0\right]$ (D) $\sin x - \sin 2x > 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$
- If $f(x) = \frac{x^2}{2 - 2 \cos x}$; $g(x) = \frac{x^2}{6x - 6 \sin x}$ where $0 < x < 1$, then
(A) 'f' is increasing function (B) 'g' is decreasing function
(C) $\frac{f(x)}{g(x)}$ is increasing function (D) $g(f(x))$ is decreasing function

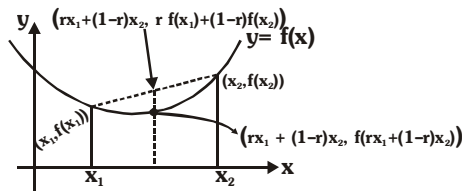
7. CONCAVITY OR CONVEXITY OF FUNCTION :

Convex function (concave upward) :

If $f(rx_1 + (1-r)x_2) \leq rf(x_1) + (1-r)f(x_2)$

$\forall x_1, x_2 \in I$ and $\forall r \in [0, 1]$

then f is said to be convex function over I (or concave upward).



Strictly Convex function (strictly concave upward) :

If $f(rx_1 + (1-r)x_2) < rf(x_1) + (1-r)f(x_2) \forall x_1, x_2 \in I$ where $x_1 \neq x_2$ and $\forall r \in (0, 1)$

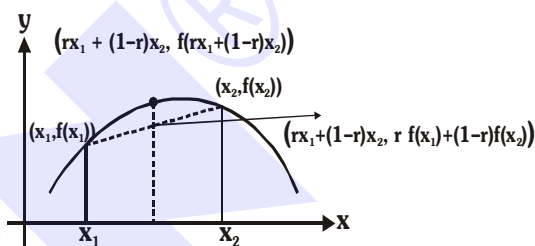
then f is said to be strictly convex function over I (or strictly concave upward).

Concave function (convex upward) :

If $f(rx_1 + (1-r)x_2) \geq rf(x_1) + (1-r)f(x_2)$

$\forall x_1, x_2 \in I$ and $\forall r \in [0, 1]$

then f is said to be concave function over I (or convex upward).



Strictly Concave function (Strictly concave upward) :

If $f(rx_1 + (1-r)x_2) > rf(x_1) + (1-r)f(x_2) \forall x_1, x_2 \in I$ where $x_1 \neq x_2$ and $\forall r \in (0, 1)$

then f is said to be Strictly concave function over I (or Strictly convex upward).

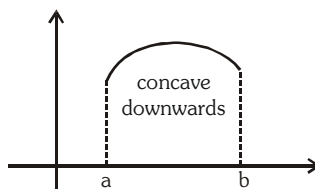
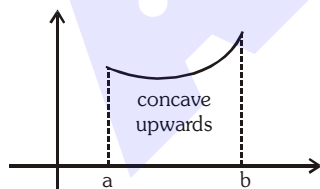
Note :

1. Straight line or linear function is said to be concave up as well as concave down.

2. The sign of the 2nd order derivative determines the concavity of the curve.

i.e. If $f''(x) \geq 0 \forall x \in (a, b)$ then graph of $f(x)$ is concave upward in (a, b) .

Similarly if $f''(x) \leq 0 \forall x \in (a, b)$ then graph of $f(x)$ is concave downward in (a, b) .



8. ROLLE'S THEOREM :

Let f be a function that satisfies the following three conditions:

- f is continuous on the closed interval $[a, b]$.
- f is differentiable on the open interval (a, b)
- $f(a) = f(b)$

Do yourself - 3 :

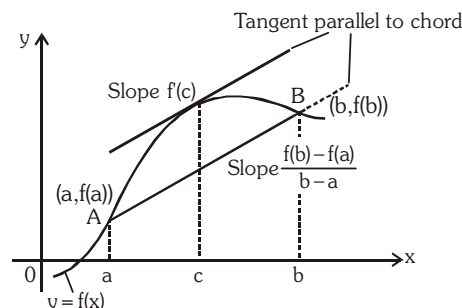
1. Verify Rolle's theorem for $y = 1 - (x^4)^{1/3}$ on the interval $[-1, 1]$
2. (a) Let $f(x) = 1 - (x^2)^{1/3}$. Show that $f(-1) = f(1)$ but there is no number c in $(-1, 1)$ such that $f'(c) = 0$. Why does this not contradict Rolle's Theorem ?
 (b) Let $f(x) = (x - 1)^{-2}$. Show that $f(0) = f(2)$ but there is no number c in $(0, 2)$ such that $f'(c) = 0$. Why does this not contradict Rolle's Theorem ?
3. If graph of $f(x) = 3x^4 + 2x^3 + ax^2 - x + 2$ is concave up for all real x , find values of a
4. Verify Rolle's theorem for the function $f(x) = x^3 - 3x^2 + 2x$ in the interval $[0, 2]$.
5. Consider the function for $x = [-2, 3]$, $f(x) = \begin{cases} \frac{x^3 - 2x^2 - 5x + 6}{x - 1} & \text{if } x \neq 1 \\ -6 & \text{if } x = 1 \end{cases}$, then
 (A) f is discontinuous at $x = 1 \Rightarrow$ Rolle's theorem is not applicable in $[-2, 3]$
 (B) $f(-2) \neq f(3) \Rightarrow$ Rolle's theorem is not applicable in $[-2, 3]$
 (C) f is not derivable in $(-2, 3) \Rightarrow$ Rolle's theorem is not applicable
 (D) Rolle's theorem is applicable as f satisfies all the conditions and c of Rolle's theorem is $1/2$
6. Exhaustive set of values of ' a ' for which the function $f(x) = x^4 + \frac{3x^2}{2} + ax^3 + 1$ will be concave upward along the entire real line, is :
 (A) $[-1, 1]$ (B) $[-2, 2]$ (C) $[0, 2]$ (D) $[0, 4]$
7. For which interval, the function $\frac{x^2 - 3x}{x - 1}$ satisfies all the conditions of Rolle's theorem
 (A) $[0, 3]$ (B) $[-3, 0]$ (C) $[1.5, 3]$ (D) For no interval
8. Rolle's theorem is not applicable to the function $f(x) = |x|$ defined on $[-1, 1]$ because
 (A) f is not continuous on $[-1, 1]$ (B) f is not differentiable on $(-1, 1)$
 (C) $f(-1) \neq f(1)$ (D) $f(-1) = f(1) \neq 0$
9. The function $f(x) = x(x + 3)e^{-(1/2)x}$ satisfies all the conditions of Rolle's theorem in $[-3, 0]$. The value of c is
 (A) 0 (B) -1 (C) -2 (D) -3

9. LAGRANGE'S MEAN VALUE THEOREM (LMVT) :

Let f be a function that satisfies the following conditions:

- (i) f is continuous in $[a, b]$
- (ii) f is differentiable in (a, b) .

Then there is a number c in (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$



(a) Geometrical Interpretation :

Geometrically, the Mean Value Theorem says that somewhere between A and B the curve has at least one tangent parallel to chord AB.

(b) Physical Interpretations :

If we think of the number $(f(b) - f(a))/(b - a)$ as the average change in f over $[a, b]$ and $f'(c)$ as an instantaneous change, then the Mean Value Theorem says that at some interior point the instantaneous change must equal the average change over the entire interval.

Illustration 12: Find c of the Lagrange's mean value theorem for the function $f(x) = 3x^2 + 5x + 7$ in the interval $[1, 3]$.

Solution : Given $f(x) = 3x^2 + 5x + 7$ (i)

$$\Rightarrow f(1) = 3 + 5 + 7 = 15 \text{ and } f(3) = 27 + 15 + 7 = 49$$

$$\text{Again } f'(x) = 6x + 5$$

$$\text{Here } a = 1, b = 3$$

Now from Lagrange's mean value theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a} \Rightarrow 6c + 5 = \frac{f(3) - f(1)}{3 - 1} = \frac{49 - 15}{2} = 17 \text{ or } c = 2.$$

Illustration 13: If $f(x)$ is continuous and differentiable over $[-2, 5]$ and $-4 \leq f'(x) \leq 3$ for all x in $(-2, 5)$, then the greatest possible value of $f(5) - f(-2)$ is -

- (A) 7 (B) 9 (C) 15 (D) 21

Solution : Apply LMVT

$$f'(x) = \frac{f(5) - f(-2)}{5 - (-2)} \text{ for some } x \text{ in } (-2, 5)$$

$$\text{Now, } -4 \leq \frac{f(5) - f(-2)}{7} \leq 3$$

$$-28 \leq f(5) - f(-2) \leq 21$$

$\therefore$ Greatest possible value of $f(5) - f(-2)$ is 21.

Do yourself - 4 :

- If $f(x) = x^2$ in $[a, b]$, then show that there exist atleast one c in (a, b) such that a, c, b are in A.P.
- Find C of LMVT for $f(x) = |x|^3$ in $[2, 5]$.
- Find c of the Lagrange's mean value theorem for the function $f(x) = 3x^2 + 5x + 7$ in the interval $[1, 3]$.
- If $f(x)$ is continuous and differentiable over $[-2, 5]$ and $-4 \leq f'(x) \leq 3$ for all x in $(-2, 5)$, then the greatest possible value of $f(5) - f(-2)$ is
(A) 7 (B) 9 (C) 15 (D) 21
- From Lagrange's mean value theorem $f(b) - f(a) = (b - a)f'(x_1)$; $a < x_1 < b$ if $f(x) = \frac{1}{x}$, then $x_1 =$
(A) $\sqrt{ab}$ (B) $\frac{a+b}{2}$ (C) $\frac{2ab}{a+b}$ (D) $\frac{b-a}{b+a}$
- Let $f(x)$ satisfy all the conditions of Lagrange's mean value theorem in $[0, 2]$. If $f(0) = 0$ and $|f'(x)| \leq \frac{1}{2}$ for all x , in $[0, 2]$ then
(A) $f(x) \leq 2$ (B) $|f(x)| \leq 1$
(C) $f(x) = 2x$ (D) $f(x) = 3$ for at least one x in $[0, 2]$
- In the Lagrange's mean value theorem $\frac{f(b) - f(a)}{b - a} = f'(c)$, if $a = 0, b = \frac{1}{2}$ and $f(x) = x(x-1)(x-2)$, the value of c is
(A) $1 - \frac{\sqrt{15}}{6}$ (B) $1 + \sqrt{15}$ (C) $1 - \frac{\sqrt{21}}{6}$ (D) $1 + \sqrt{21}$
- The abscissa of the points of the curve $y = x^3$ in the interval $[-2, 2]$, where the slope of the tangents can be obtained by Lagrange's mean value theorem for the interval $[-2, 2]$, are
(A) $\pm \frac{2}{\sqrt{3}}$ (B) $\pm \sqrt{3}$ (C) $\pm \frac{\sqrt{3}}{2}$ (D) 0
- In the Lagrange's mean value theorem, $f(b) - f(a) = (b - a)f'(c)$ if $a = 4, b = 9$ and $f(x) = \sqrt{x}$ then the value of c is
(A) 8.00 (B) 5.25 (C) 4.00 (D) 6.25

10. SPECIAL NOTE :

Use of Monotonicity in identifying the number of roots of the equation in a given interval. Suppose a and b are two real numbers such that,

- (a) Let $f(x)$ is differentiable & either strictly increasing or strictly decreasing for $a \leq x \leq b$.

&

- (b) $f(a)$ and $f(b)$ have opposite signs.

Then there is one & only one root of the equation $f(x) = 0$ in (a, b) .

MAXIMA-MINIMA

1. INTRODUCTION :

Some of the most important applications of differential calculus are optimization problems, in which we are required to find the optimal (best) way of doing something. Here are examples of such problems that we will solve in this chapter

- What is the shape of a vessel that can with-stand maximum pressure ?
- What is the maximum acceleration of a space shuttle ? (This is an important question to the astronauts who have to withstand the effects of acceleration)
- What is the radius of a contracted windpipe that expels air most rapidly during a cough ?

These problems can be reduced to finding the maximum or minimum values of a function. Let's first explain exactly what we mean by maxima and minima.

MAXIMA & MINIMA :

(a) Local Maxima/Relative maxima :

A function $f(x)$ is said to have a local maxima at $x = a$

$$\text{if } f(a) \geq f(x) \quad \forall x \in (a-h, a+h) \cap D_{f(x)}$$

Where h is some positive real number.

(b) Local Minima/Relative minima :

A function $f(x)$ is said to have a local minima

$$\text{at } x = a \text{ if } f(a) \leq f(x) \quad \forall x \in (a-h, a+h) \cap D_{f(x)}$$

Where h is some positive real number.

(c) Absolute maxima (Global maxima) :

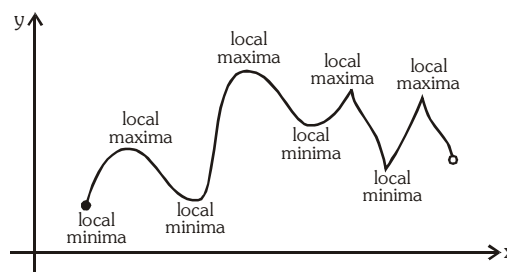
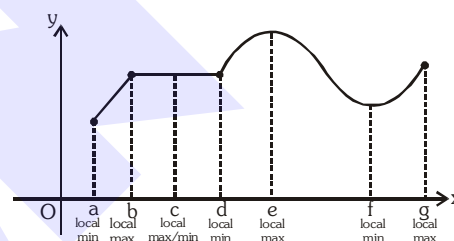
A function f has an absolute maxima (or global maxima) at c if $f(c) \geq f(x)$ for all x in D , where D is the domain of f . The number $f(c)$ is called the maximum value of f on D .

(d) Absolute minima (Global minima) :

A function f has an absolute minima at c if $f(c) \leq f(x)$ for all x in D and the number $f(c)$ is called the minimum value of f on D .

Note :

- The term 'extrema' is used for both maxima or minima.
- A local maximum (minimum) value of a function may not be the greatest (least) value in a finite interval.
- A function can have several extreme values such that local minimum value may be greater than a local maximum value.
- It is not necessary that $f(x)$ always has local maxima/minima at end points of the given interval when they are included.



Do yourself - 1 :

1. Find points of maxima or minima (if exist) for following functions

(i) $f(x) = |x| + |x - 2|$

(ii) $f(x) = |x| - |x - 2|$

(iii) $f(x) = \begin{cases} x^2 & ; -2 < x < 2 \\ x^3 & ; |x| \geq 2 \end{cases}$

2. Draw graph of $f(x) = x|x - 2|$ and, hence find points of local maxima/minima.

3. Let f be a function defined by $f(x) = \begin{cases} \frac{\tan x}{x} & , x \neq 0 \\ 1 & , x = 0 \end{cases}$

Statement - 1: $x = 0$ is point of minima of f .

Statement - 2: $f'(0) = 0$.

(1) Statement-1 is false, statement-2 is true.

(2) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement-1.

(3) Statement-1 is true, statement-2 is true; Statement-2 is not a correct explanation for statement-1.

(4) Statement-1 is true, statement-2 is false.

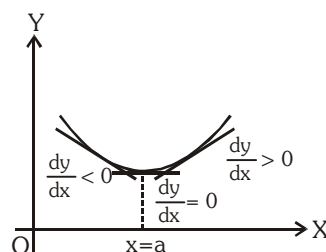
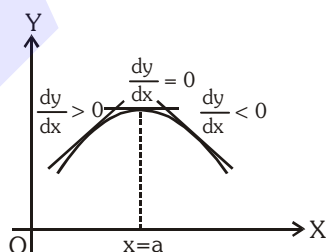
2. DERIVATIVE TEST FOR ASCERTAINING MAXIMA/MINIMA :

(a) First derivative test :

If $f'(x) = 0$ at a point (say $x = a$) and

(i) If $f'(x)$ changes sign from positive to negative in the neighbourhood of $x = a$ then $x = a$ is said to be a point **local maxima**.

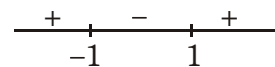
(ii) If $f'(x)$ changes sign from negative to positive in the neighbourhood of $x = a$ then $x = a$ is said to be a point **local minima**.



Note : If $f'(x)$ does not change sign i.e. has the same sign in a certain complete neighbourhood of a , then $f(x)$ is either increasing or decreasing throughout this neighbourhood implying that $x = a$ is not a point of extremum of f .

Illustration 1 : Let $f(x) = x + \frac{1}{x}$; $x \neq 0$. Discuss the local maximum and local minimum values of $f(x)$.

Solution : Here, $f'(x) = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2} = \frac{(x-1)(x+1)}{x^2}$



Using number line rule, $f(x)$ will have local maximum at $x = -1$ and local minimum at $x = 1$

$\therefore$ local maximum value of $f(x) = -2$ at $x = -1$

and local minimum value of $f(x) = 2$ at $x = 1$

Ans.

Illustration 2 : If $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \leq x \leq 2 \\ 37 - x, & 2 < x \leq 3, \end{cases}$ then

(A) $f(x)$ is increasing on $[-1, 2)$

(B) $f(x)$ is continuous on $[-1, 3]$

(C) $f'(x)$ does not exist at $x = 2$

(D) $f(x)$ has the maximum value at $x = 2$

Solution : Given, $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \leq x \leq 2 \\ 37 - x, & 2 < x \leq 3 \end{cases}$

$$\Rightarrow f'(x) = \begin{cases} 6x + 12, & -1 \leq x \leq 2 \\ -1, & 2 < x \leq 3 \end{cases}$$

(A) which shows $f'(x) > 0$ for $x \in [-1, 2)$

So, $f(x)$ is increasing on $[-1, 2)$

Hence, (A) is correct.

(B) for continuity of $f(x)$. (check at $x = 2$)

$$\text{RHL} = 35, \text{LHL} = 35 \text{ and } f(2) = 35$$

So, (B) is correct

(C) $\text{Rf}'(2) = -1$ and $\text{Lf}'(2) = 24$

so, not differentiable at $x = 2$.

Hence, (C) is correct.

(D) we know $f(x)$ is increasing on $[-1, 2)$ and decreasing on $(2, 3]$,

Thus maximum at $x = 2$,

Hence, (D) is correct.

$\therefore$ (A), (B), (C), (D) all are correct.

Ans.

Do yourself - 2 :

- Find local maxima and local minima for the function $f(x) = x^3 - 3x$.
- If function $f(x) = x^3 - 62x^2 + ax + 9$ has local maxima at $x = 1$, then find the value of a .
- Find the points of local maxima/minima of following functions -
 - $f(x) = -(x-1)^3(x+1)^2$
 - $f(x) = x \ln x$
- Find the points of local maxima/minima of function $f(x) = \operatorname{cosec}(x)$.
Hence find local maxima and local minima values of $f(x)$.
Can local maxima be less than local minima ? Justify your answer.
- Let $f(x) = \begin{cases} 3-x & 0 \leq x < 1 \\ x^2 + \ln b & x \geq 1 \end{cases}$. Find the set of values of b such that $f(x)$ has a local minima at $x = 1$.
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} k-2x, & \text{if } x \leq -1 \\ 2x+3, & \text{if } x > -1 \end{cases}$
If f has a local minimum at $x = -1$, then a possible value of k is :
 (1) 1 (2) 0 (3) $-\frac{1}{2}$ (4) -1
- Let $a, b \in \mathbb{R}$ be such that the function f given by $f(x) = \ln |x| + bx^2 + ax$, $x \neq 0$ has extreme values at $x = -1$ and $x = 2$.
Statement-1 : f has local maximum at $x = -1$ and at $x = 2$.
Statement-2 : $a = \frac{1}{2}$ and $b = \frac{-1}{4}$.
 (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.
 (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement-1.
- If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log |x| + \beta x^2 + x$ then :
 (1) $\alpha = -6$, $\beta = \frac{1}{2}$ (2) $\alpha = -6$, $\beta = -\frac{1}{2}$
 (3) $\alpha = 2$, $\beta = -\frac{1}{2}$ (4) $\alpha = 2$, $\beta = \frac{1}{2}$
- If $y = \frac{ax+b}{(x-1)(x-4)}$ has a turning value at $(2, -1)$ find a & b and show that the turning value is a maximum.
- Find the local maxima or local minima, if any, of the function $f(x) = \sin x + \cos x$, $0 < x < \frac{\pi}{2}$ using the first derivative test.
- Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ has neither maxima nor minima.

Illustration 5 : Find the global maximum and global minimum of $f(x) = \frac{e^x + e^{-x}}{2}$ in $[-\log_e 2, \log_e 7]$.

Solution : $f(x) = \frac{e^x + e^{-x}}{2}$ is differentiable at all x in its domain.

$$\text{Then } f'(x) = \frac{e^x - e^{-x}}{2}, \quad f''(x) = \frac{e^x + e^{-x}}{2}$$

$$f'(x) = 0 \Rightarrow \frac{e^x - e^{-x}}{2} = 0 \Rightarrow e^{2x} = 1 \Rightarrow x = 0$$

$$f''(0) = 1 \quad \therefore \quad x = 0 \text{ is a point of local minimum}$$

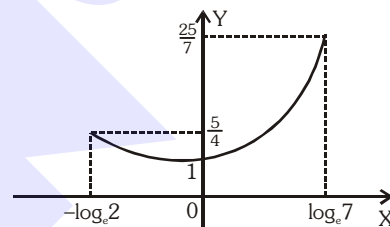
Points $x = -\log_e 2$ and $x = \log_e 7$ are extreme points.

Now, check the value of $f(x)$ at all these three points $x = -\log_e 2, 0, \log_e 7$

$$\Rightarrow f(-\log_e 2) = \frac{e^{-\log_e 2} + e^{+\log_e 2}}{2} = \frac{5}{4}$$

$$f(0) = \frac{e^0 + e^{-0}}{2} = 1$$

$$f(\log_e 7) = \frac{e^{\log_e 7} + e^{-\log_e 7}}{2} = \frac{25}{7}$$



$\therefore \quad x = 0$ is absolute minima & $x = \log_e 7$ is absolute maxima

Hence, absolute/global minimum value of $f(x)$ is 1 at $x = 0$

and absolute/global maximum value of $f(x)$ is $\frac{25}{7}$ at $x = \log_e 7$

Ans.

Illustration 6 : Identify a point of maxima/minima in $f(x) = (x + 1)^4$.

Solution : $f(x) = (x + 1)^4$
 $f'(x) = 4(x + 1)^3$

$$\begin{array}{c} \text{---ve} \quad \quad \quad \text{+ve} \\ \text{---} \quad \quad \quad \text{---} \\ \quad \quad \quad -1 \quad \quad \quad \end{array} \quad \frac{dy}{dx}$$

at $x = -1$ $f(x)$ is having local minima

$\therefore$ at $x = -1$ $f(x)$ has point of minima.

Illustration 7 : Find point of local maxima and minima of $f(x) = x^5 - 5x^4 + 5x^3 - 1$

Solution : $f(x) = x^5 - 5x^4 + 5x^3 - 1$
 $f'(x) = 5x^4 - 20x^3 + 15x^2 = 5x^2(x^2 - 4x + 3)$
 $= 5x^2(x - 1)(x - 3)$

$$f'(x) = 0 \Rightarrow x = 0, 1, 3$$

$$f''(x) = 10x(2x^2 - 6x + 3)$$

But at $x = 0$, derivative sign is positive in its neighbourhood.

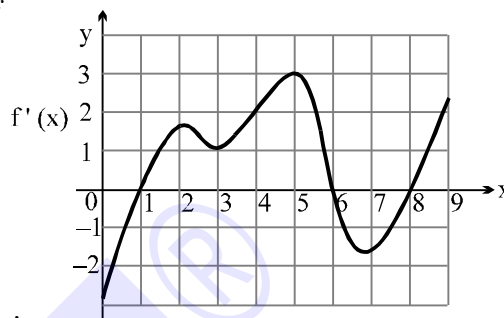
Now $f''(1) < 0 \Rightarrow$ Maxima at $x = 1$

$f''(3) > 0 \Rightarrow$ Minima at $x = 3$

$\Rightarrow$ Neither maxima nor minima at $x = 0$.

Do yourself - 3 :

1. Find local maximum value of function $f(x) = \frac{\ln x}{x}$
2. If $f(x) = x^2 e^{-2x}$ ($x > 0$), then find the local maximum value of $f(x)$.
3. Identify the point of local maxima/minima in $f(x) = (x-3)^{10}$.
4. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$
 - (i) On what intervals f is strictly increasing or strictly decreasing ?
 - (ii) At what values of x does f have a local maximum or minimum ?
 - (iii) On what intervals is $f'' > 0$ or $f'' < 0$
 - (iv) State the x -coordinate(s) of the point(s) of inflection.
 - (v) Assuming that $f(0) = 0$, sketch a graph of f .
5. The function $f(x)$ defined for all real numbers x has the following properties
 $f(0) = 0$, $f(2) = 2$ and $f'(x) = k(2x - x^2)e^{-x}$ for some constant $k > 0$. Find
 - (a) the intervals on which f is strictly increasing and strictly decreasing and any local maximum or minimum values.
 - (b) the intervals on which the graph f is concave down and concave up.
 - (c) the function $f(x)$ and plot its graph.



3. USEFUL FORMULAE OF MENSURATION TO REMEMBER :

- (a) Volume of a cuboid = ℓbh .
- (b) Surface area of a cuboid = $2(\ell b + bh + h\ell)$.
- (c) Volume of a prism = area of the base $\times$ height.
- (d) Lateral surface area of prism = perimeter of the base $\times$ height.
- (e) Total surface area of a prism = lateral surface area + 2 area of the base
(Note that lateral surfaces of a prism are all rectangles).
- (f) Volume of a pyramid = $\frac{1}{3}$ area of the base $\times$ height.
- (g) Curved surface area of a pyramid = $\frac{1}{2}$ (perimeter of the base) $\times$ slant height.
(Note that slant surfaces of a pyramid are triangles).
- (h) Volume of a cone = $\frac{1}{3} \pi r^2 h$.
- (i) Curved surface area of a cylinder = $2 \pi rh$.
- (j) Total surface area of a cylinder = $2 \pi rh + 2 \pi r^2$.
- (k) Volume of a sphere = $\frac{4}{3} \pi r^3$.
- (l) Surface area of a sphere = $4 \pi r^2$.
- (m) Area of a circular sector = $\frac{1}{2} r^2 \theta$, when θ is in radians.

4. SUMMARY OF WORKING RULES FOR SOLVING REAL LIFE OPTIMIZATION PROBLEM :

- First :** When possible, draw a figure to illustrate the problem & label those parts that are important in the problem. Constants & variables should be clearly distinguished.
- Second :** Write an equation for the quantity that is to be maximized or minimized. If this quantity is denoted by 'y', it must be expressed in terms of a single independent variable x. This may require some algebraic manipulations.
- Third :** If $y=f(x)$ is a quantity to be maximum or minimum, find those values of x for which $dy/dx=f'(x)=0$.
- Fourth :** Using derivative test, test each value of x for which $f'(x)=0$ to determine whether it provides a maximum or minimum or neither.
- Fifth :** If the derivative fails to exist at some point, examine this point as possible maximum or minimum.
- Sixth :** If the function $y=f(x)$ is defined only for $x \in [a,b]$ then examine $x=a$ & $x=b$ for possible extreme values.

Illustration 8 : Determine the largest area of the rectangle whose base is on the x-axis and two of its vertices lie on the curve $y = e^{-x^2}$.

Solution :

Area of the rectangle will be $A = 2a \cdot e^{-a^2}$

$$\text{For max. area, } \frac{dA}{da} = \frac{d}{da}(2ae^{-a^2}) = e^{-a^2}[2 - 4a^2]$$

$$\frac{dA}{da} = 0 \Rightarrow a = \pm \frac{1}{\sqrt{2}}$$

& sign of $\frac{dA}{da}$ changes from positive to negative at $a = +\frac{1}{\sqrt{2}}$

$$\Rightarrow x = \frac{1}{\sqrt{2}} \text{ are points of maxima} \Rightarrow A_{\max} = \frac{2}{\sqrt{2}} \cdot e^{-\left(\frac{1}{\sqrt{2}}\right)^2} = \frac{\sqrt{2}}{e^{1/2}} \text{ sq units.} \quad \text{Ans.}$$

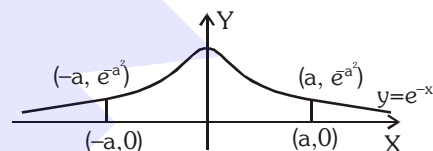
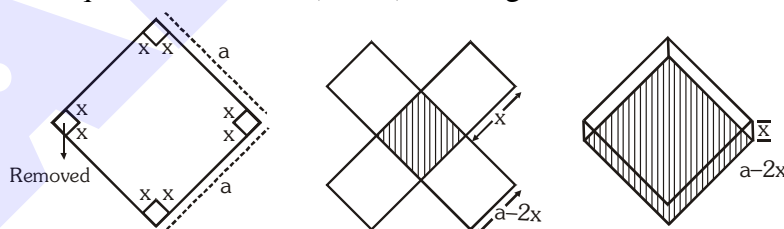


Illustration 9 : A box of maximum volume with top open is to be made by cutting out four equal squares from four corners of a square tin sheet of side length a ft, and then folding up the flaps. Find the side of the square base cut off.

Solution : Volume of the box is, $V = x(a - 2x)^2$ i.e., squares of side x are cut out then we will get a box with a square base of side $(a - 2x)$ and height x.



$$\therefore \frac{dV}{dx} = (a - 2x)^2 + x \cdot 2(a - 2x)(-2)$$

$$\frac{dV}{dx} = (a - 2x)(a - 6x)$$

$$\text{For } V \text{ to be extremum } \frac{dV}{dx} = 0 \Rightarrow x = a/2, a/6$$

But when $x = a/2$; $V = 0$ (minimum) and we know minimum and maximum occurs alternately in a continuous function.

Hence, V is maximum when $x = a/6$.

Ans.

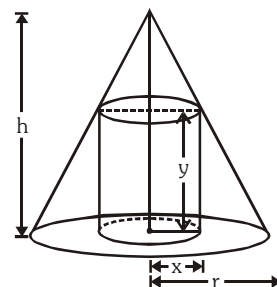
Illustration 10 : If a right circular cylinder is inscribed in a given cone. Find the dimension of the cylinder such that its volume is maximum.

Solution : Let x be the radius of cylinder and y be its height

$$V = \pi x^2 y$$

x, y can be related by using similar triangles

$$\frac{y}{r-x} = \frac{h}{r} \Rightarrow y = \frac{h}{r}(r-x)$$



$$\Rightarrow V(x) = \pi x^2 \frac{h}{r}(r-x) \Rightarrow V(x) = \frac{\pi h}{r}(rx^2 - x^3)$$

$$V'(x) = \frac{\pi h}{r}(2rx - 3x^2)$$

$$V'(x) = 0 \Rightarrow x = 0, \frac{2r}{3}$$

$$V''(x) = \frac{\pi h}{r}(2r - 6x)$$

$$V''(0) = 2\pi h \Rightarrow x = 0 \text{ is point of minima}$$

$$V''\left(\frac{2r}{3}\right) = -2\pi h \Rightarrow x = \frac{2r}{3} \text{ is point of maxima}$$

Thus volume is maximum at $x = \left(\frac{2r}{3}\right)$ and $y = \frac{h}{3}$.

Illustration 11 : Let a cuboid having square base has area 6. Then find its maximum volume.

Solution : Total area = $2a^2 + 4ah = 6$

$$h = \frac{(6-2a^2)}{4a} \Rightarrow V = a^2 h = \frac{a^2(6-2a^2)}{4a}$$

$$\frac{dv}{da} = 6 - 6a^2 = 0$$

$$a = 1 \quad \& \quad \frac{d^2v}{da^2} = -ve$$

Maximum $V = 1$

Do yourself - 4 :

- Find the two positive numbers x & y such that their sum is 60 and xy^3 is maximum.
- If from a wire of length 36 metre, a rectangle of greatest area is made, then find its two adjacent sides in metre.
- If $ab = 2a + 3b$ where $a > 0$, $b > 0$, then find the minimum value of ab .
- Of all closed right circular cylinders of a given volume of 100 cubic centimetres, find the dimensions of cylinder which has minimum surface area.
- A wire of length 2 units is cut into two parts which are bent respectively to form a square of side $= x$ units and a circle of radius $= r$ units. If the sum of the areas of the square and the circle so formed is minimum, then :
 (1) $2x = r$ (2) $2x = (\pi + 4)r$ (3) $(4 - \pi)x = \pi r$ (4) $x = 2r$
- A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is :
 (1) $9/2$ (2) $9/7$ (3) $7/9$ (4) $2/9$
- The plan view of a swimming pool consists of a semicircle of radius r attached to a rectangle of length ' $2r$ ' and width ' s '. If the surface area A of the pool is fixed, for what value of ' r ' and ' s ' the perimeter ' P ' of the pool is minimum.
- A closed rectangular box with a square base is to be made to contain 1000 cubic feet. The cost of the material per square foot for the bottom is 15 paise, for the top 25 paise and for the sides 20 paise. The labour charges for making the box are Rs. 3/-. Find the dimensions of the box when the cost is minimum.
- Find the maximum perimeter of a triangle on a given base ' a ' and having the given vertical angle α .
- A cylinder is obtained by revolving a rectangle about the x -axis, the base of the rectangle lying on the x -axis and the entire rectangle lying in the region between the curve $y = \frac{x}{x^2 + 1}$ & the x -axis.
 Find the maximum possible volume of the cylinder.

Important note :

- (i) If the sum of two real numbers x and y is constant then their product is maximum if they are equal.

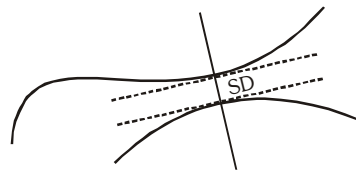
$$\text{i.e. } xy = \frac{1}{4} [(x+y)^2 - (x-y)^2]$$

- (ii) If the product of two positive numbers is constant then their sum is least if they are equal.

$$\text{i.e. } (x+y)^2 = (x-y)^2 + 4xy$$

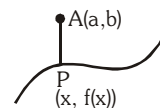
5. LEAST/GREATEST DISTANCE BETWEEN TWO CURVES :

Least/Greatest distance between two non-intersecting curves usually lies along the common normal. (Wherever defined)



Note : Given a fixed point $A(a, b)$ and a moving point $P(x, f(x))$ on the curve $y = f(x)$. Then

AP will be maximum or minimum if it is normal to the curve at P .



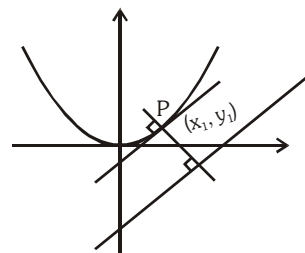
Proof : $F(x) = (x - a)^2 + (f(x) - b)^2 \Rightarrow F'(x) = 2(x - a) + 2(f(x) - b) \cdot f'(x)$

$$\therefore f'(x) = -\left(\frac{x - a}{f(x) - b}\right). \text{ Also } m_{AP} = \frac{f(x) - b}{x - a}. \text{ Hence } f'(x) \cdot m_{AP} = 1.$$

Illustration 12 : Find the co-ordinates of the point on the curve $x^2 = 4y$, which is at least distance from the line $y = x - 4$.

Solution : Let $P(x_1, y_1)$ be a point on the curve $x^2 = 4y$ at which normal is also a perpendicular to the line $y = x - 4$.

$$\text{Slope of the tangent at } (x_1, y_1) \text{ is } 2x = 4 \frac{dy}{dx} \Rightarrow \frac{dy}{dx} \Big|_{(x_1, y_1)} = \frac{x_1}{2}$$



$$\therefore \frac{x_1}{2} = 1 \Rightarrow x_1 = 2$$

$$\therefore x_1^2 = 4y_1 \Rightarrow y_1 = 1$$

Hence required point is $(2, 1)$

Do yourself - 5 :

- Find the coordinates of point on the curve $y^2 = 8x$, which is at minimum distance from the line $x + y = -2$.
- The shortest distance between line $y - x = 1$ and curve $x = y^2$ is :-

(1) $\frac{8}{3\sqrt{2}}$

(2) $\frac{4}{\sqrt{3}}$

(3) $\frac{\sqrt{3}}{4}$

(4) $\frac{3\sqrt{2}}{8}$

- A perpendicular is drawn from the centre to a tangent to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Find the greatest value of the intercept between the point of contact and the foot of the perpendicular.

6. SOME SPECIAL POINTS ON A CURVE :

(a) **Stationary points:** The stationary points are the points of domain where $f'(x) = 0$.

(b) **Critical points :** There are three kinds of critical points as follows :

- The point at which $f'(x) = 0$
- The point at which $f'(x)$ does not exist
- The end points of interval (if included)

These points belong to domain of the function.

Note : Local maxima and local minima occurs at critical points only but not all critical points will correspond to local maxima/local minima.

(c) **Point of inflection :** A point where the graph of a function has a tangent line and where the strict concavity changes is called a point of inflection. For finding point of inflection of any function,

compute the points (x-coordinate) where $\frac{d^2y}{dx^2} = 0$ or $\frac{d^2y}{dx^2}$ does

not exist. Let the solution is $x = a$, if $\frac{d^2y}{dx^2} = 0$ at

$x = a$ and sign of $\frac{d^2y}{dx^2}$ changes about this point then it is called point of inflection.

if $\frac{d^2y}{dx^2}$ does not exist at $x = a$ and sign of $\frac{d^2y}{dx^2}$ changes about this point and tangent exist at this point then it is called point of inflection.

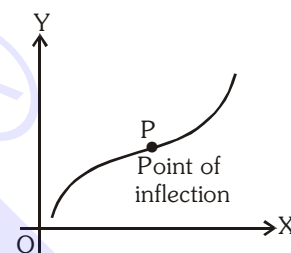


Illustration 13 : Find the critical point(s) & stationary point(s) of the function $f(x) = (x-2)^{2/3}(2x+1)$

Solution : $f(x) = (x-2)^{2/3}(2x+1)$

$$f'(x) = (x-2)^{2/3} \cdot 2 + (2x+1) \cdot \frac{2}{3}(x-2)^{-1/3} = 2(x-2)^{2/3} + \frac{2}{3}(2x+1) \frac{1}{(x-2)^{1/3}}$$

$$= \left[2(x-2) + \frac{2}{3}(2x+1) \right] \frac{1}{(x-2)^{1/3}} = \frac{2(5x-5)}{3(x-2)^{1/3}}$$

$f'(x)$ does not exist at $x = 2$ and $f'(x) = 0$ at $x = 1$

$\therefore x = 1, 2$ are critical points and $x = 1$ is stationary point.

Illustration 14 : The point of inflection for the curve $y = x^{5/3}$ is -

- (A) (1, 1) (B) (0, 0) (C) (1, 0) (D) (0, 1)

Solution : Here $\frac{d^2y}{dx^2} = \frac{10}{9x^{1/3}}$

From the given points we find that (0, 0) is the point of the curve where

$\frac{d^2y}{dx^2}$ does not exist but sign of $\frac{d^2y}{dx^2}$ changes about this point.

$\therefore (0, 0)$ is the required point

Ans. (B)

Illustration 15 : Find the inflection point of $f(x) = 3x^4 - 4x^3$. Also draw the graph of $f(x)$ giving due importance to maxima, minima and concavity.

Solution :

$$f(x) = 3x^4 - 4x^3$$

$$f'(x) = 12x^3 - 12x^2$$

$$f'(x) = 12x^2(x - 1)$$

$$f'(x) = 0 \Rightarrow x = 0, 1$$

examining sign change of $f'(x)$

thus $x = 1$ is a point of local minima

$$f''(x) = 12(3x^2 - 2x)$$

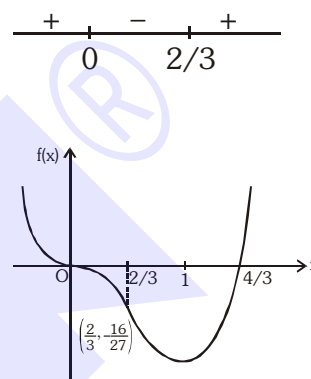
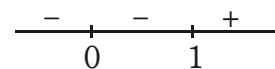
$$f''(x) = 12x(3x - 2)$$

$$f''(x) = 0 \Rightarrow x = 0, \frac{2}{3}$$

Again examining sign of $f''(x)$

thus $x = 0, \frac{2}{3}$ are the inflection points

Hence the graph of $f(x)$ is



Do yourself - 6 :

- Find the critical points and stationary point of the function $f(x) = \frac{e^x}{x}$
- Find the point of inflection for the curve $y = x^3 - 6x^2 + 12x + 5$
- For the function $f(x) = \frac{x^4}{12} - \frac{5x^3}{6} + 3x^2 + 7$
 - Find the interval in which $f''(x) > 0$
 - Find the interval in which $f''(x) < 0$
 - Find the points of inflection of $f(x)$.